



Sensory Marketing Measures in an Algerian Beauty Clinic: An Exploratory Pre–Post Study

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Abstract: This study explored the association between sensory marketing measures and client experience in an Algerian beauty clinic. This study adopted a descriptive-analytical approach, based on a questionnaire administered to a convenience sample of 30 female clients of the clinic. Both descriptive and inferential statistical analyses were conducted to test the research hypotheses. The findings revealed that the overall application of sensory marketing in the clinic was positively associated with client experience. The pre–post comparison showed a statistically significant difference in overall client experience between the two measurement phases. Following the intervention, significant associations were observed between the sensory stimuli (sight, sound, smell, taste, and touch) and client experience. Among these five dimensions, sound showed the strongest association, followed by sight and taste. This research offered practical insights into how sensory marketing could be leveraged as a tool for enhancing client experience. Understanding changes in clients’ perceptions could help inform service improvements and sensory-marketing decisions.

Keywords: Algeria; Beauty clinic; Client experience; Exploratory pre–post study; Sensory marketing

1 Introduction

For decades, marketing has been viewed as a rational process that emerged in response to consumer needs. Consumers were then believed to make their choices based on product characteristics and the benefits offered, as well as on concrete factors such as price, features, and utility. This perspective was reflected in traditional models of consumer behavior, such as the Howard-Sheth model, which conceptualized purchasing decisions as information-processing and evaluative processes [1]. However, recent studies have shown that the buying process is not as rational as initially assumed: when evaluating brands, consumers primarily rely on their emotions (shaped by personal experiences and current feelings) rather than on objective information such as facts, features, or brand attributes [2–4].

Sensory marketing is a recently established phenomenon; it has transformed consumer experience. Whereas marketing communication was once largely monologic, it has become increasingly interactive and multidimensional, with brands finding their voices and consumers often responding in subconscious ways [5, 6]. Early sensory marketing efforts relied mainly on visual elements in posters and billboards, including variations in color and typography. Over time, marketers have expanded their attention to other sensory dimensions and have begun to recognize the effectiveness of using certain scents to influence consumers, resulting in increased sales for some businesses [7, 8]. Nowadays, sensory marketing is employed by a large number of companies across various sectors such as food, cosmetics, hotels, spas, bakeries, and coffee shops [9–11]. This shift towards sensory marketing taps into the human senses (sight, sound, smell, taste, and touch) to create immersive experiences that influence perception and decision making at a subconscious level. Unlike traditional marketing, which predominantly engages cognitive processes, sensory marketing influences the emotional and instinctual parts of the brain, leading to deeper brand associations and longer-term consumer loyalty [9].

This study aims to examine how sensory marketing is associated with variations in perceived client experience within an Algerian beauty clinic setting called “WASA”. Beauty clinics, which provide the specific context and contribution of this study, remain underexplored in the sensory-marketing literature. Prior studies on sensory

marketing focused on various sectors; therefore, this research addressed this gap by extending the literature to this specific service context.

Established in 2022 in Cheraga, Algiers, the WASA Clinic specializes in dermatological care, aesthetic medical procedures, and anti-aging treatments. It offers women a range of services, including filler injections, laser treatments, skin rejuvenation, and other related services. This choice reflects a broader trend in which even beauty clinics are no longer viewed as sterile and medical spaces. Instead, they are evolving into luxurious wellness destinations where clients seek not just treatment but an entire experience that appeals to their emotions and senses. This approach has been widely adopted in beauty clinics and sensory-driven elements are carefully designed to enhance client satisfaction and encourage repeat visits. Accordingly, this study examined how these sensory and service-related measures were associated with changes in clients' perceptions and experiences.

The research question was formulated as follows: *Is there a significant difference in client experience before and after the implementation of sensory marketing measures at WASA Clinic?* The objective is to investigate whether the inclusion of sensory components in the WASA Clinic was associated with significant differences in the overall perception of service quality, comfort, and client satisfaction, and to identify which specific sensory dimensions (sight, sound, smell, taste, and touch) showed the strongest associations with perceived client experience.

2 Conceptual Background and Development of Hypotheses

2.1 Sensory Marketing

In the mid-20th century, marketers began recognizing the limitations of conventional advertising, which primarily focused on visual and auditory stimuli. Cheskin [12] introduced the concept of “*sensation transference*”, demonstrating that consumers' perceptions of the value and quality of a product could be significantly influenced by its sensory attributes, such as packaging color and texture. His work laid the foundation for understanding how sensory cues impacted consumer behavior. While the senses of taste, touch, and sight have been part of marketing research for a longer time, the term sensory marketing emerged primarily with the inclusion of the other senses, namely smell and sound.

As markets became increasingly saturated, the need to differentiate products led to the exploration of multi-sensory engagement. Researchers delved into how integrating stimuli across sight, sound, smell, taste, and touch could enhance brand perception and loyalty. This interdisciplinary approach combined insights from psychology, neuroscience, and marketing to craft strategies that appeal to consumers' subconscious desires and emotions. For customers to perceive their interactions with a brand as meaningful, they should be able to emotionally connect to the brand. Sensory marketing is the strategic use of the five senses to influence consumers' perception, evoke emotions, and enhance brand experiences. According to Krishna [9], sensory marketing was “*the process of engaging consumers' senses and affecting their perception, judgment, and behavior*”. Her study highlighted how sensory cues such as colors, textures, and sounds could shape consumers' emotions, so they would perceive products as more luxurious, comforting, or exciting. One compelling study [13] demonstrated how tactile experiences influenced perception of products. The researchers found that when consumers could physically touch a product (e.g., holding a soft fabric or a sturdy object), their perception of its quality and value increased significantly compared to when they only saw it online or in advertisements. This study supported the idea that physical interaction with products could drive purchasing decisions at an instinctual level.

At the psychological level, sensory marketing works because human senses are linked to the limbic system, the part of the brain responsible for emotions and memory. Research has shown that sensory stimuli could evoke subconscious memories and associations, influencing consumer behavior without them realizing it. Indeed, a previous study [14] emphasized that scent had a powerful link to memory and emotions. Research by North and Hargreaves [15] reported that background music in a store could influence shoppers' moods, shopping pace, and even perception of product pricing. Furthermore, Labrecque and Milne [16] showed that color schemes in branding and store environments significantly affected consumers' emotions.

2.2 Client Experience

Client experience has emerged as a key concept in marketing, focusing on the creation of unique, enjoyable, and memorable experiences [17]. Holbrook and Hirschman [18] described client experience as involving various playful leisure activities, sensory pleasures, daydreams, esthetic enjoyment, and emotional responses. Client value is therefore created by the total experience of all elements [19] rather than by any single element. Client experience refers to the sum of all interactions a client has with a brand throughout their entire journey, from initial awareness to post-purchase engagement. It encompasses both functional and emotional aspects of a customer's relationship with a company, thus influencing their perception, satisfaction, and loyalty. Unlike customer service, which focuses on direct support interactions, client experience is a holistic concept that includes marketing, sales, service delivery, digital interactions, and even word-of-mouth influences. Meyer and Schwager [20] defined client experience as “*the internal and subjective response that customers have to any direct or indirect contact with a company*”. This means

client experience is not just about delivering a good product or service; it is about how clients feel during and after every interaction with a brand. A well-crafted client experience strategy ensures that every touchpoint is optimized to meet customers' needs, fostering long-term loyalty and advocacy.

In the present study, we used the term *client experience* rather than *consumer experience*, as we consider it more appropriate for the context of a beauty clinic. Customers' perceptions and interactions are central to shaping the overall client experience. Understanding and optimizing these key factors not only enhances client satisfaction but also provides a strategic advantage in building stronger relationships with clients [21]. The literature identified several elements that contributed to improving customer experience, such as personalization [22], ease of interaction [23], service quality [24], sensory engagement [9], feedback, and continuous improvement [25].

2.3 Sensory Stimuli and Their Role in Shaping Client Experience

Sensory marketing is a technique that aims to “*seduce consumers by engaging their senses in order to influence their feelings and behaviour*” [26]. Several researchers have examined the impact of the five senses or more broadly, sensory stimuli, on client experience.

The first sense, sight, was described as the most seductive sense [27] and the dominant sensory modality [9, 28]. Sight enables individuals to perceive and make sense of the physical environment as it allows them to detect changes and differences in their surroundings [29]. This sense is influenced by the intensity of light, with research showing that such stimulation affects behaviour, feelings, and well-being [9, 30]. The literature has mainly focused on vision as a stimulus in marketing [31, 32], with notable applications in packaging, advertising, point-of-sale design, and website interfaces. Moreover, the literature suggested its dominance in client experiences over other senses, such as touch and taste [9, 33].

The second sense, sound, influences consumers' moods, behaviors, and preferences [9, 34, 35]. It also shapes the perception of product quality and affects the period consumers remain in stores [36]. The sense of hearing has important implications for customer experience, particularly in relation to the amount of time spent in the store, as well as the pace and movement within the retail environment [9].

The third sense, smell, has strong connections to memory, as demonstrated by neuroscience research [9, 33, 37]. Goldkuhl and Styvén [38] noted that the sense of smell, while intricately connected to emotions and memories, was related to pleasure and well-being. Smell evokes more emotional memories compared to those generated by other types of sensory stimulation [14]. Likewise, it provides more detailed autobiographical memories [39] and older ones than those formed through auditory and visual information [40]. The memories of smells exhibit a flatter forgetting curve over time compared to those acquired through other sensory modalities [9, 33]. With the help of scent stimulation, it is possible to induce consumer behavior and influence their purchasing decisions.

The penultimate sense, taste, is the least explored dimension in sensory marketing, and its management has primarily been applied in the field of gastronomy. Research has established that clients could distinguish five basic tastes: sweet, salty, sour, bitter, and umami, a taste often described as delicious or savory [41]. The literature acknowledged that each perception of taste resulted from the interplay of all five human senses. It is not straightforward to identify a taste using this sense alone, as taste perception depends on the integration of multiple sensory inputs [9]. Consequently, taste should be managed by marketing practitioners from an integrative rather than a singular perspective when designing consumer experience.

The final sense, touch, is among the first sensory modalities to develop during prenatal life [42] and the last to decline with age [9]. It remains the least explored within marketing research [43]. Touch is often regarded as the most intimate of the senses, as it conveys information about the material characteristics of objects, such as texture, softness, temperature, and weight [44], all of which contribute to the initial perception of product quality [45]. In the context of sensory marketing, the sense of touch has been recognized as influencing client purchasing behavior through physical interaction with products [46]. Research indicated that consumers increasingly based their purchasing decisions on tactile experiences; physical interaction with products could stimulate impulse buying and effectively increase the quantity of products purchased [43]. This tendency capitalizes on clients' inherent need to handle products [47], as they prefer to assess and gather information about a product's attributes using their hands and fingers [46]. Moreover, tactile contact has been shown to facilitate memory of the product [32].

2.4 Hypotheses and Research Model

Based on the above conceptual framework and literature review, three hypotheses were formulated at various levels of analysis:

H_1 (Global level): *The overall application of sensory marketing in the clinic is positively associated with client experience.*

H_2 (Dimensional level): *Each sensory dimension (sight, sound, smell, taste, and touch) is positively associated with client experience.*

H_3 (Temporal level): *Client experience differs significantly between the pre- and post-implementation phases.*

Our conceptual model is presented in Figure 1. In this model, three pathways for enhancing client experience were proposed. The first pathway hypothesized a positive change in client experience as a result of implementing sensory marketing (H_1). The second pathway hypothesized positive variations in client experience associated with each individual sensory stimulus (H_2). The third pathway hypothesized positive differences in client experience between the two phases (before–after) (H_3).

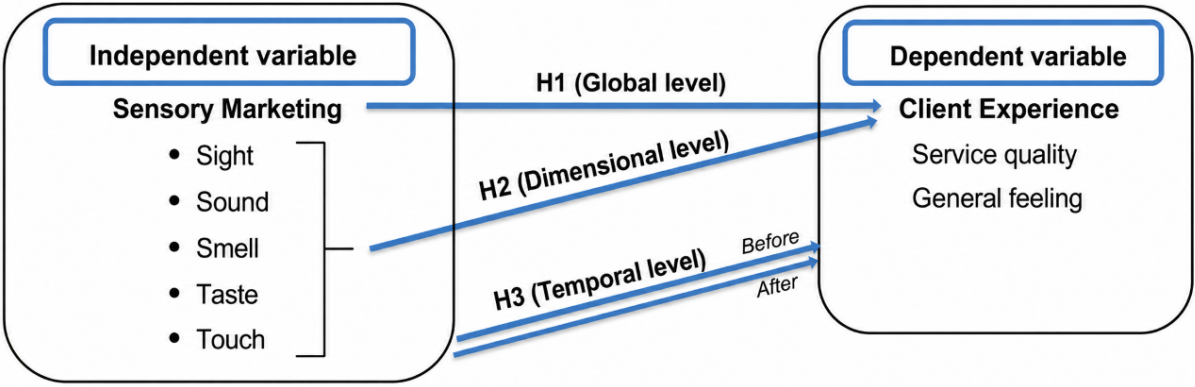


Figure 1. Research model

3 Methodology

3.1 Participants

As WASA Clinic primarily serves women, this study was conducted on a convenience sample of 30 female participants. Participants were identified by the clinic manager based on his knowledge of regular clients who had previous experience with the clinic’s services. All selected participants ($n = 30$) provided informed consent to participate in the study. Regarding age distribution, 11 participants were aged 18–24 years, 13 were aged 25–34 years, and 6 were aged 35–44 years. For the objectives of this exploratory study, other demographic characteristics were considered not essential.

As the clinic serves clients comes from various administrative regions (wilayas) across the country, who expect services to be provided within a relatively brief time, this time constraint partly explains the small size of the sample. Indeed, 30 clients agreed to devote their time to completing the questionnaire and did so twice. However, given the exploratory nature of this study, which does not aim to generalize findings, this sample size was considered acceptable. The same 30 participants completed the questionnaire before and after the questionnaire before and after the implementation of sensory marketing.

3.2 Research Variables and Improvement Measures

Table 1 presents the study variables, including sensory marketing as the independent variable, represented by its five sensory stimuli, and client experience as the dependent variable, comprising two dimensions: service quality and general feeling. The table also lists the implemented improvement measures, along with their classification.

As presented in Table 1, the intervention focused on enhancing existing sensory elements rather than introducing new components. The service-process co-interventions consisted only of modest adjustments to existing practices. All improvement measures were implemented simultaneously between March and April 2025. Clinic staff actively participated in the implementation of these improvements.

All women who agreed to participate completed the questionnaire. Given the small sample size, the researcher administered the questionnaires personally by asking the questions directly to the respondents. To minimize potential interviewer bias, the researcher maintained a neutral attitude throughout the data collection process and followed a standardized procedure by providing the same instructions to all participants. Participants were assured of the confidentiality of their responses and were encouraged to provide honest evaluations of their experiences. Furthermore, clinic staff were not present during the administration of questionnaires, hence enabling participants to provide their opinions freely.

The first evaluation was conducted at the beginning of March, prior to the implementation of the various measures. Between the two phases, WASA Clinic implemented improvements to certain aspects of the environment and services, as indicated in Table 1. The participants completed the same questionnaires again at the end of April to assess changes in the client experience. The clinic’s pricing structure remained unchanged throughout the

intervention period to control potential price-related confounding effects on clients' perceptions and the observed changes in client experience.

Table 1. Research variables and improvement measures

Variables		Improvement Measures	Intervention Classification
Independent variable— Sensory marketing	Sight	Use of lighter and more harmonious colors; improved lighting; enhanced interior decoration; improved cleanliness and spatial organization; optimization of humidity control and air quality	Sensory-environment intervention
	Sound	Enhancement of soft background music; creation of a pleasant sound environment	
	Touch	Improved seating comfort; enhanced material quality; enhancement of tactile elements; improvement of temperature control and humidity regulation	
	Smell	Enhancement of the olfactory environment through pleasant scents	
	Taste	Improvement of refreshment services; enhancement of the gustatory experience	
Dependent variable— Client experience	Service quality	Enhanced reception service; improved appointment scheduling process; reduced waiting time; clearer explanations provided	Service-process co-intervention
	General feeling	Improvement of overall perception and clinic image	

3.3 Measuring Instruments

The measurement items were developed based on a literature review on sensory marketing, incorporating both theoretical and empirical contributions relating to the five sensory dimensions [9, 32]. The questionnaire items were not directly adopted item-by-item from previous studies. Instead, they were developed based on key sensory and service-related elements identified in the literature (e.g., background music, interior decoration, materials, tactile elements, scents, refreshments, etc.). The items were further formulated by the authors to reflect the specific context of a beauty clinic.

The same questionnaire was administered twice: before and after the implementation of sensory marketing measures at the WASA Clinic. It was initially administered in French, as the clinic's clientele was predominantly French-speaking, and was subsequently translated into English for the purpose of this research. To match pre- and post-intervention responses while preserving confidentiality, participants were assigned unique identification codes ranging from 1 to 30. They were instructed to use the same code when completing both questionnaires and this arrangement allowed the researchers to link responses across the two measurement points without collecting personal identifying information.

The questionnaire was divided into two parts. The first part included 18 questions related to sensory marketing distributed across the five stimuli (5 for sight, 3 for sound, 4 for touch, 4 for smell, and 2 for taste), while the second part included 10 questions related to the client experience distributed across two dimensions (5 for service quality and 5 for general feeling). The instrument was structured using a three-point Likert scale, where respondents indicated their level of agreement with various statements: 1 = Strongly Disagree, 2 = Somewhat Agree, and 3 = Strongly Agree. This simplified scale with three alternative responses was adopted to differentiate three levels of agreement (low, moderate, and high) rather than to assess attitudes with greater precision through a scale incorporating a larger number of response categories and a neutral midpoint. The objective was to make the questionnaire more straightforward and accessible for participants, thus minimizing the risk of confusion or indecision. This approach is particularly appropriate in contexts where participants may have limited time or varying familiarity with survey instruments, thereby enhancing response quality and completion rates [48].

Given the exploratory nature of this study, no formal pilot testing was conducted prior to data collection. However, the authors reviewed the items to ensure their clarity, relevance, and alignment with the research objectives.

3.4 Data Analysis and Hypothesis Testing

Data were analyzed using both descriptive and inferential statistics. Descriptive statistics were used to summarize the characteristics of the study variables, while inferential statistics were employed to test the proposed hypotheses.

To test H_1 , *Spearman's rank correlation coefficient* was adopted to examine the positive association between the overall application of sensory marketing and client experience. For H_2 , *Spearman's rank correlation coefficient* was also employed to assess whether each sensory marketing dimension was positively associated with client experience. Finally, H_3 was assessed using the *Wilcoxon signed-rank test* to determine whether significantly positive differences in client experience existed between the pre-implementation and post-implementation phases.

4 Results

To assess the internal consistency of the measurement scales used in this study, Cronbach's α was calculated in Table 2.

Table 2. Reliability test of internal consistency (Cronbach's α)

Variables	Number of Questions	α Before Implementation	α After Implementation
Sensory marketing	18	0.889	0.858
Client experience	10	0.902	0.740

Note: $n = 30$.

The obtained Cronbach's α values exceeded the commonly accepted threshold of 0.7, indicating satisfactory internal consistency among the items within each overall scale.

4.1 Descriptive Analysis

The mean sensory-marketing score increased from 2.64 to 2.89, underscoring an overall higher level of agreement and satisfaction. The standard deviation (*SD*) decreased from 0.356 to 0.181, suggesting greater consistency in respondents' positive evaluations. For the dependent variable client experience, the mean score increased from 2.73 to 2.93, and the *SD* decreased from 0.309 to 0.132. These results shown in Table 3 indicate a notable improvement in perceived service quality following the sensory marketing intervention.

Table 3. Pre–post descriptive statistics of the two variables

Variables	Mean		SD	
	Before	After	Before	After
Sensory marketing	2.64	2.89	0.356	0.181
Client experience	2.73	2.93	0.309	0.132

Note: $n = 30$; Three-point Likert scale: 1 = Strongly Disagree, 2 = Somewhat Agree, 3 = Strongly Agree.

SD = Standard deviation. Overall scores were calculated as the unweighted mean of the dimension scores.

Table 4 sets out the descriptive statistics (means and standard deviations) for the dimensions of sensory marketing and client experience (The percentages of agreement for the different items are presented in Appendix Table A1).

The results presented in Appendix Table A1 confirmed a consistently elevated level of agreement with the visual aspects of the clinic both before and after the sensory marketing intervention, with a notably positive shift in perception across most items. The perception of the clinic's colors improved slightly, with the proportion of respondents who strongly agreed that the colors created a calming and professional atmosphere increasing from 86.7% to 93.3%. Lighting was also perceived more positively after the intervention, with strong agreement rising from 83.3% to 90%, indicating a greater contribution to relaxation. The already well-appreciated interior decoration gained even stronger approval from 80% to 93.3% of the respondents, hence reinforcing the perception of quality. Cleanliness and organization showed an increase in strong agreement, rising from 86.7% to 96.7%, representing the highest score within this dimension though these aspects were already strong attributes of the clinic. A notable improvement was observed in the perception of air quality, when strong agreement increased from 70% to 86.7% following the installation of dehumidifiers, thus suggesting an enhanced perception of environmental comfort. For sight stimuli, the mean score increased from 2.77 to 2.91, indicating a higher overall level of agreement and satisfaction. The standard deviation decreased from 0.375 to 0.276, reflecting greater consistency in respondents' positive evaluations following the changes.

Based on percentage shifts and mean scores before and after the intervention, the clinic's cozy environment also showed marked improvement. A clear upward trend in positive perception and auditory comfort was recorded across sound-related items. The percentage of participants who strongly agreed that background music created a relaxing

atmosphere increased from 36.7% to 63.3%. Similarly, the strong agreement that the overall sound environment was pleasant rose from 63.3% to 93.3%, suggesting enhanced appreciation of the acoustic atmosphere. Perceptions of the contribution of the new sound system increased from 60% to 86.7%, indicating a more favorable evaluation of the audio environment. For sound stimuli, the mean score increased from 2.43 to 2.79, while the *SD* decreased from 0.518 to 0.344, implying more consistent and uniformly positive responses.

The results also indicated improvement in the perception of smell-related factors following the intervention. Strong agreement that a pleasant scent was noticeable at the entrance of the clinic increased from 73.3% to 96.7%. The perception that scented candles or diffusers enhanced the sensory welcome rose from 60% to 86.7%. Agreement that ambient fragrance reinforced the impression of cleanliness increased from 76.7% to 96.7%. The statement regarding scent contributing to a feeling of luxury increased from 90% to 93.3%. The mean score for olfactory stimuli rose from 2.70 to 2.90, while the *SD* decreased from 0.396 to 0.305, indicating stronger and more consistent agreement after the changes.

Table 4. Pre-post descriptive statistics of the different dimensions

Variables	Dimensions	Before		After	
		Mean	<i>SD</i>	Mean	<i>SD</i>
Sensory marketing	Sight	2.77	0.375	2.91	0.276
	Sound	2.43	0.518	2.79	0.344
	Smell	2.70	0.396	2.90	0.305
	Touch	2.69	0.448	2.90	0.192
	Taste	2.63	0.600	2.98	0.091
Client experience	Service quality	2.69	0.399	2.91	0.154
	General feeling	2.77	0.479	2.96	0.110

Note: $n = 30$; Three-point Likert scale: 1 = Strongly Disagree, 2 = Somewhat Agree, 3 = Strongly Agree. The scores represent the mean values of items included in each construct; *SD* = Standard deviation.

Improvements were observed in clients' tactile experience. The strong agreement that seats in the waiting room were comfortable increased from 80% to 100%. The perception that materials were pleasant to the touch rose from 76.7% to 96.7%. The agreement that tactile elements helped clients relax increased from 70% to 80%. The proportion of participants who perceived temperature and humidity as pleasant rose from 73.3% to 90%. The mean score for touch stimuli increased from 2.69 to 2.90, while the *SD* decreased from 0.448 to 0.192, reflecting more consistently positive evaluations.

Taste-related sensory aspects also showed improvement after the intervention. The perception that gustatory attention contributes to comfort during the visit increased from 73.3% to 100%. Agreement that refreshments contribute positively to the overall experience rose from 76.7% to 96.7%. The mean score for taste stimuli increased from 2.63 to 2.98, while the *SD* decreased from 0.600 to 0.091, indicating stronger agreement and greater response consistency following the changes.

Descriptively, the increase in mean scores and percentages following the intervention is consistent with the hypothesis (H_1) that higher sensory-marketing ratings were associated with higher client-experience ratings.

Finally, the results indicated an improvement in perceived service quality and general feeling following the service-process co-interventions. Regarding perceived service quality, the mean score increased from 2.69 to 2.91, while the standard deviation decreased from 0.399 to 0.154. For general feeling, the mean score increased from 2.77 to 2.96, while the standard deviation decreased from 0.479 to 0.110. These figures suggested an increase in the overall level of agreement and satisfaction, with more consistent responses among participants.

4.2 Statistical Analysis

Data distribution was assessed using both the Kolmogorov–Smirnov and Shapiro–Wilk normality tests (Table 5), which confirmed non-normality.

4.2.1 Spearman correlation analysis before intervention

This analysis aims to examine the strength and direction of the associations between key variables. This correlation matrix (Table 6) examined the relationship between total sensory marketing (before the intervention) and client experience.

Table 6 indicates no statistically significant correlation between sensory marketing and client experience before the intervention ($\rho = 0.180$, $p = 0.342$). However, the analysis of the underlying dimensions showed that sensory

marketing was significantly associated with perceived service quality ($\rho = 0.479, p = 0.007$), whereas no significant association was observed with general feeling ($\rho = 0.267, p = 0.154$).

Table 5. Kolmogorov–Smirnov and Shapiro–Wilk normality test results

Dimensions	Before Implementation				After Implementation			
	K–S Statistic	<i>p</i> Value	S–W Statistic	<i>p</i> Value	K–S Statistic	<i>p</i> Value	S–W Statistic	<i>p</i> Value
Sight	0.408	0.001	0.412	<0.001	0.333	0.002	0.679	0.002
Sound	0.363	<0.001	0.675	0.001	0.240	0.001	0.870	0.002
Smell	0.495	0.001	0.377	<0.001	0.250	0.002	0.751	<0.001
Touch	0.432	<0.001	0.591	<0.001	0.287	0.001	0.735	0.003
Taste	0.539	0.002	0.180	<0.001	0.396	<0.001	0.654	<0.001
Service quality	0.412	<0.001	0.624	<0.001	0.221	0.001	0.770	<0.001
General feeling	0.508	<0.001	0.411	<0.001	0.410	<0.001	0.542	<0.001

Note: $n = 30$; K–S = Kolmogorov–Smirnov statistic; S–W = Shapiro–Wilk statistic.

Table 6. Pre-intervention correlations between sensory marketing and client experience

Variables	Sensory Marketing	Client Experience
Sensory marketing	1	0.180 ($p = 0.342$)
Client experience	0.180 ($p = 0.342$)	1

Note: $n = 30$; ρ = Spearman's correlation coefficient. These correlation analyses were exploratory and should be interpreted with caution.

Table 7 presents the correlations between the different sensory dimensions (sight, sound, smell, touch, and taste) and the overall client experience before the implementation of sensory marketing improvements. The aim is to examine the extent to which sensory elements were initially associated with clients' perceptions of service quality and satisfaction.

Table 7. Pre-intervention correlations between sensory dimensions and client experience

Variables	Client Experience	Sight	Sound	Smell	Touch	Taste
Client experience	1	0.021 ($p = 0.911$)	0.251 ($p = 0.180$)	0.131 ($p = 0.490$)	0.007 ($p = 0.970$)	−0.022 ($p = 0.909$)
Sight	-	1	0.505** ($p = 0.004$)	0.072 ($p = 0.705$)	0.634** ($p < 0.001$)	−0.102 ($p = 0.593$)
Sound	-	-	1	0.159 ($p = 0.401$)	0.855** ($p < 0.001$)	−0.137 ($p = 0.469$)
Smell	-	-	-	1	0.063 ($p = 0.739$)	0.454* ($p = 0.012$)
Touch	-	-	-	-	1	−0.111 ($p = 0.560$)
Taste	-	-	-	-	-	1

Note: $n = 30$; ρ = Spearman's correlation coefficient. **denotes $p < 0.01$ and *denotes $p < 0.05$, indicating a significant correlation. “-” indicates not applicable.

The correlation analysis conducted prior to the introduction of sensory marketing provided interesting insights into the interrelationships between the five sensory dimensions and the overall client experience in the clinic. While none of the individual senses of sight, sound, smell, touch, or taste was significantly associated with the overall client experience, there were strong interrelationships among the sensory variables themselves. Specifically, vision was strongly correlated with both sound ($\rho = 0.505, p = 0.004$) and touch ($\rho = 0.634, p < 0.001$), and sound was especially strongly correlated with touch ($\rho = 0.855, p < 0.001$). Clients who were more aware of visual or auditory elements tended to report higher sensitivity to tactile aspects such as comfort, textures, or the physical ambiance. Furthermore, there was a moderately positive association between gustatory and olfactory perceptions ($\rho = 0.454$,

$p = 0.012$), thus reflecting their inherently sensory connection and upholding the idea that these two modalities frequently interacted in shaping perceptual experience.

Although no directly significant association with the overall experience was observed, the strong inter-sensory correlations suggested potentially synergistic relationships among sensory elements if the environment was optimized. This finding underscored the value of adopting a holistic sensory approach in marketing strategies rather than addressing individual senses in isolation.

4.2.2 Spearman correlation analysis after intervention

This correlation matrix (Table 8) examined the relationship between sensory marketing (after the intervention) and client experience.

Table 8 indicates a strong, positive, and statistically significant association ($\rho = 0.638$, $p < 0.001$) between adoption of sensory marketing and client experience after its implementation. This result indicated that clients who reported higher awareness of or engagement with sensory elements, including scent, sound, and visual aspects also reported higher ratings of their experiences. The strength of this association suggested an obvious correspondence between improvements in sensory elements and clients' evaluations of service quality and satisfaction.

Therefore, this finding provided empirical support for the first hypothesis (H_1), which posits that the overall application of sensory marketing in the clinic is positively associated with client experience.

Table 8. Post-intervention correlations between sensory marketing and client experience

Variables	Sensory Marketing	Client Experience
Sensory marketing	1	0.638** ($p < 0.001$)
Client experience	0.638** ($p < 0.001$)	1

Note: $n = 30$; ρ = Spearman's correlation coefficient. **denotes $p < 0.01$, indicating a significant correlation.

Table 9 illustrates the correlations between client experience and the five sensory stimuli following the implementation of sensory marketing measures. The aim is to assess the strength and statistical significance of the associations between each sensory dimension and the overall perceived experience.

Table 9. Post-intervention correlations between sensory dimensions and client experience

Variables	Client Experience	Sight	Sound	Smell	Touch	Taste
Client Experience	1	0.525** ($p = 0.003$)	0.587** ($p = 0.001$)	0.362* ($p = 0.049$)	0.487** ($p = 0.006$)	0.521** ($p = 0.003$)
Sight	-	1	0.489** ($p = 0.006$)	0.620** ($p < 0.001$)	0.497** ($p = 0.005$)	0.474** ($p = 0.008$)
Sound	-	-	1	0.502** ($p = 0.005$)	0.519** ($p = 0.003$)	0.382* ($p = 0.037$)
Smell	-	-	-	1	0.484** ($p = 0.007$)	0.524** ($p = 0.003$)
Touch	-	-	-	-	1	0.420* ($p = 0.021$)
Taste	-	-	-	-	-	1

Note: $n = 30$; ρ = Spearman's correlation coefficient. **denotes $p < 0.01$ and *denotes $p < 0.05$, indicating a significant correlation. "-" indicates not applicable.

Table 9 reflected the associations between the five sensory dimensions and overall client experience following the implementation of sensory marketing measures in the clinic. The highest correlation coefficients were observed for auditory stimulation ($\rho = 0.587$, $p = 0.001$) and visual stimulation ($\rho = 0.525$, $p = 0.003$) in relation to overall client experience. Clients who reported higher awareness or responsiveness to auditory and visual elements also tended to report higher evaluations of their experience. Both touch ($\rho = 0.487$, $p = 0.006$) and taste ($\rho = 0.521$, $p = 0.003$) showed statistically significant positive associations, suggesting that comfort-related features, such as cushioned seating, tactile elements, or provisions for refreshments were associated with higher ratings of client experience. Perception of smell ($\rho = 0.362$, $p = 0.049$), the weakest among the five senses, displayed a significantly positive association. This indicated that olfactory elements, such as essential oils or scented candles, were associated with more favorable perceptions of the clinic's atmosphere, though to a lesser extent than other sensory dimensions. Overall, these results highlighted the cumulative associations of multiple sensory dimensions: clients' experiences appeared to reflect the associations across several sensory dimensions rather than a single dominant factor. This

finding provided empirical support for hypothesis (H_2), which posits that each sensory dimension is positively associated with client experience.

4.2.3 Pre-post comparison of sensory marketing changes

Given the paired ordinal nature of the data, the three-point response scale, the small sample size, and the observed ceiling effects, a non-parametric analysis using the Wilcoxon signed-rank test was adopted for the pre-post comparison. This test was used to compare clients' responses before and after the sensory marketing changes. This non-parametric test helped determine whether the differences observed in paired data were statistically significant. It was applied to both sensory perceptions and client experience (Table 10).

Table 10. Wilcoxon signed-rank test results

Statistic	Sensory Marketing	Client Experience
Positive ranks (before < after)	21	23
Mean positive rank	14.69	17.50
Sum of positive ranks	308.5	402.5
Negative ranks (before > after)	6	6
Mean negative rank	11.58	5.42
Sum of negative ranks	69.5	32.5
Ties (after = before)	3	1
Total participants	30	30
Z statistic	-2.871	-4.003
Effect size (r)	0.52	0.73
p Value	0.004	<0.001

Note: $n = 30$; Effect size (r) = $|Z|/\sqrt{n}$.

Wilcoxon signed-rank test scores (Table 10) revealed significantly positive changes in participants' perceptions of sensory marketing following the implementation of the multisensory improvements. Twenty-one participants (70%) reported higher post-intervention scores, whereas six participants (20%) reported lower scores and three reported no changes. This difference was statistically significant ($Z = -2.871$, $p = 0.004$, $r = 0.52$), indicating a substantial increase in sensory marketing ratings between the two measurement phases. Similarly, overall client experience scores increased significantly following the intervention. Twenty-three participants (76.7%) reported higher scores, six reported lower scores, and one reported no changes. The Wilcoxon test confirmed that this difference was statistically significant ($Z = -4.003$, $p < 0.001$, $r = 0.73$), suggesting a substantial increase in overall client experience ratings after the implementation of the improvements.

Taken together, these results indicated statistically significant improvements in reported client experience and perceptions of sensory marketing, supporting hypothesis (H_3) regarding positive differences between the pre- and post-implementation phases.

5 Discussion

Based on the results of the current research, it could be concluded that the proposed general hypothesis was supported.

First, the descriptive analysis indicated increases in positive responses across multiple dimensions of sensory marketing and client experience, after the intervention (Table 4). Sound-related items showed the largest descriptive increase in favorable responses, consistent with earlier studies [9, 34, 35]. Similarly, taste-related elements exhibited increases in favorable responses. Touch-related aspects received prominent levels of agreement from participants, indicating that tactile elements were positively perceived and consistent with previous research [13, 46]. Positive changes were also observed for olfactory aspects, suggesting a potential contribution to a more pleasant sensory environment [38]. Overall, improvements were found in visual ambiance, sound environment, scent, tactile comfort, and perceived service quality, implying that the implemented sensory marketing measures were associated with more favorable perceptions of the clinic's atmosphere and service-related aspects. These findings are consistent with previous literature emphasizing the role of visual cues in shaping perceptions of professionalism, cleanliness, and comfort [31, 32]. The increase in "Strongly Agree" responses after the intervention (Appendix Table A1) indicates a more favorable evaluation of the clinic environment and service quality, aligning with previous findings that sensory stimulation was associated with enhanced perceptions of service environments [9].

Beyond these descriptive changes, the correlation analysis revealed a strong positive and statistically significant association between sensory marketing adoption and client experience after the intervention (Table 8). Together, these observations provided support for Hypothesis (H_1), which proposes that implementation of sensory marketing would be associated with more positive client perceptions and experiences in the WASA beauty clinic.

Second, the correlation analysis further illustrated the associations between the different sensory dimensions and overall client experience after the intervention (Table 9). The results showed positive associations between sensory dimensions and client experience, with sound, sight, and taste exhibiting the strongest observed relationships. These findings support the second hypothesis (H_2), which proposes a positive association between sensory stimuli and perceived client experience. Following the intervention, significant associations were observed between sensory dimensions and client experience. This suggests that the sensory factors examined were positively related to participants' overall perceptions of the clinic environment and service experience. These findings are consistent with previous research highlighting the role of multisensory experiences in shaping perceived service quality [9, 32]. However, these findings should be interpreted as associations rather than causal effects.

Finally, the last finding indicates that sensory marketing was associated with changes in client experience, supporting hypothesis (H_3). This observation was supported by the statistically significant results of the Wilcoxon signed-rank test (Table 10). The increase in positive responses suggests that clients perceived the clinic as a more pleasant, memorable, and welcoming environment. These findings are consistent with Hultén's perspective that sensory experiences could contribute to stronger emotional connections with places [32].

Overall, the findings emphasized the prominent role of sensory marketing in shaping clients' perceptions and experiences, and highlighted its association with a strengthened client-clinic relationship, hence providing quantitative evidence consistent with the research hypotheses and theoretical expectations.

6 Limitations of the Study

This study has several limitations that should be considered when interpreting the findings. First, the one-group pre-post design allowed only the observation of variations and associations between two measurement points and did not support causal conclusions regarding the effects of sensory stimuli. Accordingly, the differences observed, particularly for sound, should be interpreted as variations between the pre- and post-intervention phases rather than evidence of a causal effect or the intrinsic superiority of any specific sensory stimulus. Second, the sensory interventions were implemented alongside improvement measures of service quality within the clinic. As a result, the observed changes in clients' perceptions might not be attributable solely to sensory marketing, as these complementary initiatives might have influenced the outcomes.

Third, although the questionnaire items were developed based on relevant literature, no formal assessment of content validity was conducted. Moreover, both sensory marketing perceptions and client experience were measured through self-reported responses collected using the same instrument, which might have increased the risk of common-method bias. In addition, the use of a three-point response scale might have reduced response variability and increased the likelihood of ceiling effects, particularly for items that received fairly high ratings before the intervention.

Another limitation concerns the sample. The study was conducted in a single clinic with a relatively small number of participants, which restricts the generalizability of the findings. Furthermore, participants were recruited through manager-assisted convenience sampling among regular clients of the clinic. This approach may have introduced selection bias and affected the representativeness of the sample.

Finally, although correlation coefficients were reported before and after the intervention, no formal statistical tests were performed to determine whether the observed differences between these correlations were statistically significant.

Taken together, these limitations call for caution when interpreting the results. They highlight the need for future studies to consider these methodological constraints to appropriately isolate the effects of sensory stimuli and further examine their associations with client experience.

7 Conclusions

This exploratory study examined how implementing measures to stimulate the five senses was associated with variations in the client experience. Using an empirical quantitative approach, the research observed that visual ambiance, calming music, pleasant scents, tactile comfort, and subtle gustatory elements were linked to more favorable overall perceptions at the WASA Clinic. These patterns were reflected not only in descriptive statistics and increases in favorable-response percentages, but were also supported by inferential analyses, including the Wilcoxon signed-rank test and Spearman's correlation coefficients.

This study provided evidence consistent with the three hypotheses proposed in the research model: first, that the implementation of sensory marketing is associated with more positive client perceptions; second, that there are positive associations between sensory stimuli and client experience; and third, that observed differences between the two phases (before and after the intervention) were associated with more favorable clients' perceptions.

The findings demonstrated that, among the five dimensions of sensory marketing, sound showed the strongest association with client experience, followed by sight and taste. However, future studies are encouraged to enlarge the sample size and conduct more advanced analyses to further examine the predictive associations of sensory marketing

dimensions on client experience. A larger sample would also enhance the robustness of the findings and contribute to the generalizability of the results across different service contexts.

The practical implications of this research remain substantial, as it offers insights and recommendations for clinic managers and wellness professionals seeking to enhance the client experience through intentional sensory strategies. When effectively aligned with a brand's identity, sensory stimuli could contribute to creating a more engaging environment and might support stronger client satisfaction and loyalty. The findings indicated that auditory and visual dimensions might represent key areas for managerial consideration, as they demonstrated the greatest observed changes between the pre- and post-intervention phases and the strongest associations with overall client experience. Consequently, in contexts where resources are constrained, clinic managers may initially prioritize the optimization of auditory and visual environments through initiatives such as the selection of suitable background music, improvement of acoustic comfort, enhancement of interior design, and development of a visually consistent atmosphere. Nevertheless, these findings should be interpreted cautiously, as correlation coefficients reflect the strength of associations rather than causal relationships. Therefore, strategic decisions regarding sensory investments should take into account implementation costs, organizational priorities, and the broader service context.

Author Contributions

Conceptualization, B.A.R.; methodology, B.A.R.; software, B.A.R.; validation, B.W.; formal analysis, B.A.R. and B.W.; investigation, B.A.R.; resources, B.A.R.; data curation, B.A.R.; writing—original draft preparation, B.W.; writing—review and editing, B.W.; visualization, B.W.; supervision, B.W.; project administration, B.A.R. and B.W. All authors have read and agreed to the published version of the manuscript.

Data Availability

This study formed part of a master's thesis in Entrepreneurship and Project Management conducted under an agreement between the National Higher School of Management and WASA Clinic. A copy of the thesis is held in the library of the National Higher School of Management. The supporting data are available from the corresponding author upon reasonable request.

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Informed Consent Statement

All participants provided informed consent before participating in the study. They were told the purpose of the research, the voluntary nature of their participation, and the confidentiality of their responses.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this work, the authors used AI-assisted technologies to support language editing, structural refinement, and clarity of expression. The authors reviewed, verified, and edited all outputs and took full responsibility for the content of the submitted manuscript.

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Appendix

Table A1. Percentage of agreement for each questionnaire item

Dimensions	Items	Before			After		
		Strongly Disagree	Somewhat Agree	Strongly Agree	Strongly Disagree	Somewhat Agree	Strongly Agree
Sight	The clinic's colors create a calming and professional atmosphere	6.7%	6.7%	86.7%	—	6.7%	93.3%
	The lighting contributes to a feeling of relaxation	3.3%	13.3%	83.3%	—	10%	90%
	The interior decoration reinforces the image of quality	3.3%	16.7%	80%	—	6.7%	93.3%
	Cleanliness and organization inspire confidence	3.3%	10%	86.7%	—	3.3%	96.7%
	Dehumidifiers improve air quality	6.7%	23.3%	70%	6.7%	6.7%	86.7%
Sound	The background music creates a relaxing atmosphere	6.7%	56.7%	36.7%	6.7%	30%	63.3%
	The overall sound environment (noise level, calmness, etc.) is pleasant	10%	26.7%	63.3%	—	6.7%	93.3%
	The sound system creates an ambiance	13.3%	26.7%	60%	—	13.3%	86.7%
	A pleasant scent is noticeable at the clinic entrance	3.3%	23.3%	73.3%	—	3.3%	96.7%
Smell	Scented candles or diffusers improve the sensory welcome	10%	30%	60%	10%	3.3%	86.7%
	The ambient fragrance reinforces the impression of cleanliness	3.3%	20%	76.7%	—	3.3%	96.7%
	The scent gives a feeling of luxury	3.3%	6.7%	90%	3.3%	3.3%	93.3%
Touch	The seats in the waiting room are comfortable	—	20%	80%	—	—	100%
	The materials used (tables, walls, and chairs) are pleasant to touch	3.3%	20%	76.7%	—	3.3%	96.7%
	Tactile objects (toys, head massagers, etc.) help to relax	10%	20%	70%	6.7%	13.3%	80%
	The temperature and humidity in the clinic are pleasant	10%	16.7%	73.3%	—	10%	90%
Taste	Gustatory elements (e.g., drinks and snacks) enhance comfort during the visit	10%	16.7%	73.3%	—	—	100%
	The refreshments contribute to a positive experience	13.3%	10%	76.7%	—	3.3%	96.7%
Service quality	The reception staff demonstrated professionalism and warmth	3.3%	10%	86.7%	—	—	100%
	The process of appointment scheduling was efficient and timely	6.7%	23.3%	70%	—	6.7%	93.3%
	The waiting time prior to service delivery was appropriate	13.3%	36.7%	50%	6.7%	20%	73.3%
	The staff provided clear explanations regarding the treatments offered	3.3%	6.7%	90%	—	3.3%	96.7%
	The overall organization of the service facilitated my visit	3.3%	16.7%	80%	—	—	100%

Continued on next page

Table A1 (continued)

Dimensions	Items	Before			After		
		Strongly Disagree	Somewhat Agree	Strongly Agree	Strongly Disagree	Somewhat Agree	Strongly Agree
General feeling	I felt comfortable throughout my visit	3.3%	16.7%	80%	—	6.7%	93.3%
	I perceived the clinic to be professional	3.3%	16.7%	80%	—	—	100%
	The overall atmosphere encourages my return	3.3%	16.7%	80%	—	6.7%	93.3%
	I would recommend this clinic to others	10%	3.3%	86.7%	—	—	100%
	My experience met my expectations	6.7%	6.7%	86.7%	—	6.7%	93.3%

Note: $n = 30$; Three-point Likert scale: 1 = Strongly Disagree, 2 = Somewhat Agree, and 3 = Strongly Agree. “—” indicates 0%.